

Requirements

This document lists what PrintXpress must do (functional requirements) and the qualities it must have (non-functional requirements). Each requirement has an id so it can be traced to user stories, diagrams, and test cases.

Functional requirements

Authentication and profile

Id	Requirement	Stories
FR-01	The app shall let a customer register with a full name, an email or phone number, and a password.	US-01
FR-02	The app shall reject a register attempt that uses an email or phone already in the database.	US-01, US-03
FR-03	The app shall hash every password before saving it, and never store plain text.	US-01
FR-04	The app shall let a customer log in with their email or phone and password.	US-02
FR-05	The app shall keep a customer logged in until they log out.	US-02, US-05
FR-06	The app shall let a customer view and edit their name, email, and phone.	US-03

FR-07	The app shall let a customer add, edit, delete, and set a default delivery address.	US-04
FR-08	The app shall let a customer log out, which clears the login state.	US-05

Browsing

Id	Requirement	Stories
FR-09	The app shall show product categories loaded from the local database.	US-06
FR-10	The app shall show a list of products for a chosen category.	US-06
FR-11	The app shall show product details including name, description, base price, material, sample image, and the size, material, and quantity options.	US-07
FR-12	The app shall let a customer filter or search the product list by type, material, or size.	US-08

Ordering

Id	Requirement	Stories
FR-13	The app shall let a customer upload an image or PDF design file and show the chosen file name.	US-09

FR-14	The app shall let a customer enter custom text in place of a file.	US-10
FR-15	The app shall require either a file or custom text before an item is added to an order.	US-09, US-10
FR-16	The app shall let a customer choose quantity, material or paper, and size, and shall update the price to match.	US-11
FR-17	The app shall let a customer choose pickup or delivery, and select a saved address for delivery.	US-11, US-04
FR-18	The app shall save the order and its items to the database with the status set to processing.	US-11
FR-19	The app shall show an order confirmation with the order number and a summary.	US-11
FR-20	The app shall let a customer save a design with a title and reuse it on a later order.	US-12

Order management

Id	Requirement	Stories
FR-21	The app shall list a customer's orders with date, total, and status.	US-13
FR-22	The app shall show order status as one of processing, printing, ready for pickup, out for delivery, completed, or cancelled.	US-13

FR-23	The app shall allow cancel and reschedule only while the order status is processing.	US-14
FR-24	The app shall hide or disable cancel and reschedule once the status is printing or later.	US-14

Notifications and offers

Id	Requirement	Stories
FR-25	The app shall create a notification when an order is placed and when it is completed.	US-15
FR-26	The app shall show notifications in a list with an unread marker, and mark one read when it is opened.	US-15
FR-27	The app shall show active seasonal offers with title, discount, code, and valid dates, and hide expired offers.	US-16

Help and support

Id	Requirement	Stories
FR-28	The app shall show print guidelines covering file types, resolution, colour mode, and bleed.	US-17
FR-29	The app shall show an FAQ list and the shop's contact details.	US-18

Non-functional requirements

Id	Quality	Requirement
NFR-01	Platform	The app shall run on Android 7.0 (API 24) and above, built with native Kotlin and Jetpack Compose only.
NFR-02	Offline	The app shall work fully offline. All data is stored on the device with Room over SQLite.
NFR-03	Performance	Common screens shall open in under one second on a mid-range phone, and lists shall scroll smoothly.
NFR-04	Usability	The UI shall use one accent colour on a white background with dark gray text, consistent labels, and sentence case throughout.
NFR-05	Accessibility	Text shall meet a readable contrast against white, touch targets shall be at least 48 by 48 dp, and every icon button shall have a content description.
NFR-06	Security	Passwords shall be hashed. No sensitive data shall be logged or shown in plain text.
NFR-07	Reliability	A failed action, such as a duplicate email or a missing file, shall show a clear message and shall not crash the app.
NFR-08	Validation	All input forms shall validate required fields and formats before saving.
NFR-09	Maintainability	Code shall be grouped by purpose (data, ui, viewmodel), use simple patterns, and carry short comments.

NFR-10	Portability	The database shall be a single local SQLite file that can be exported and opened with DB Browser for SQLite.
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Validation rules

These rules back FR-02, FR-15, FR-16, and NFR-08.

1. Full name is required and at least two characters.
2. Email, when used, matches a basic email pattern. Phone, when used, is digits only and 9 to 12 characters long.
3. Password is at least six characters.
4. An order item needs a file or custom text, a quantity of at least one, and a selected size and material.
5. A delivery order needs a selected address.